

EXTREME NETWORKS

Universal Hardware USB conversion to Fabric Engine

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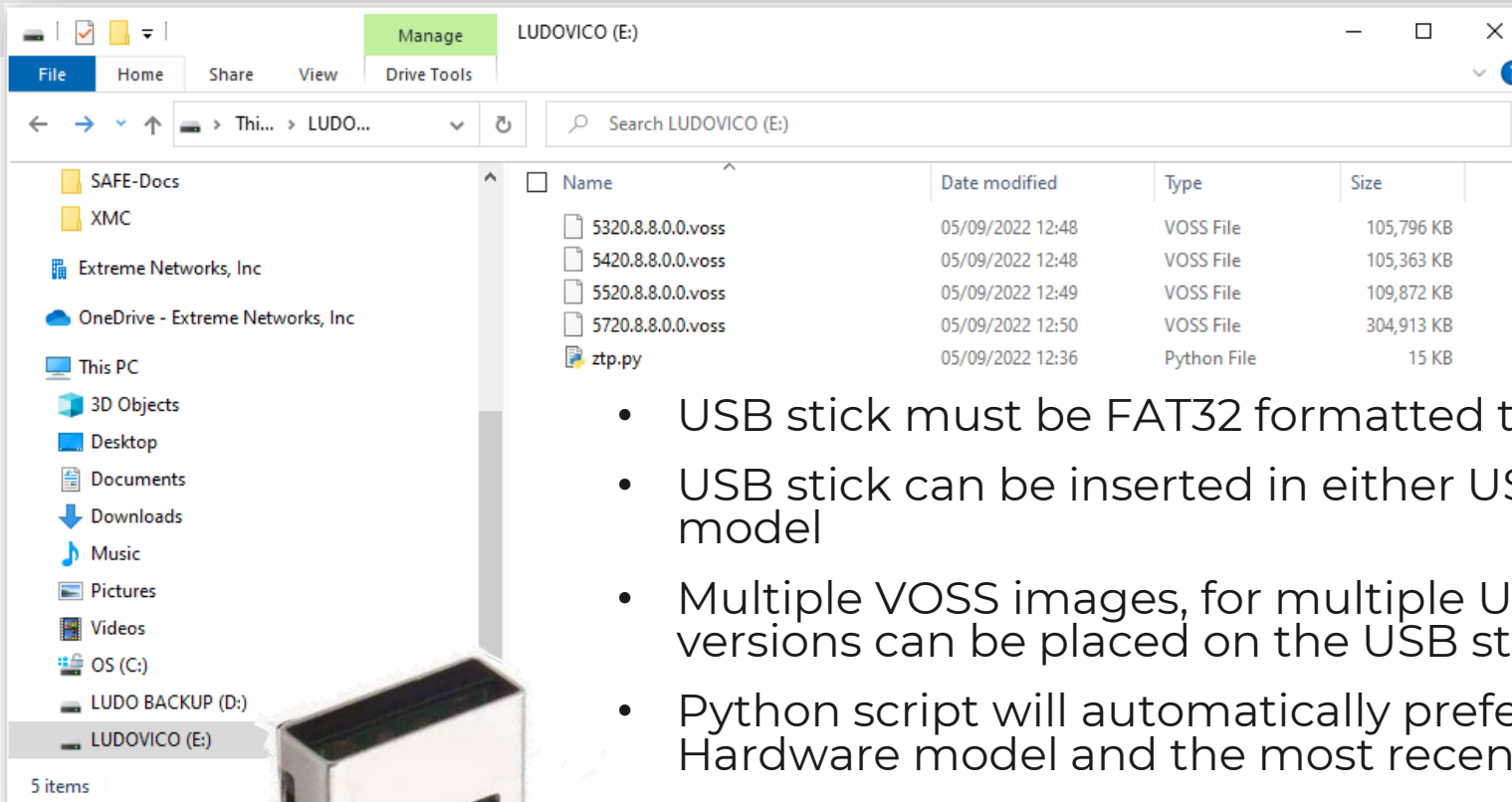
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Converting Universal Hardware to Fabric Engine via USB stick



- **Universal Hardware: 7720/7520/5720/5520/5420/5320/4220...**
- **USB stick with files:**
 - `ztp.py` : Python script from Github; must be named “ztp.py”
<https://github.com/extremenetworks/ExtremeScripting/tree/master/EXOS/Python/ztp-convert-to-fabric-engine>
 - `<uni-hw-image>.voss` : VOSS image file(s) for Universal hardware
- **Brand new switch, out of the box:**
 1. Insert USB stick
 2. Power on the switch
- **Switch already booted into EXOS as Switch Engine**
 1. Insert USB stick
 2. Execute CLI command: `unconfigure switch`

USB stick VOSS images



- USB stick must be FAT32 formatted to be recognized by the switch
- USB stick can be inserted in either USB port 1 or port 2, depending on switch model
- Multiple VOSS images, for multiple Universal Hardware models and VOSS versions can be placed on the USB stick
- Python script will automatically prefer the image for the correct Universal Hardware model and the most recent software version
- VOSS conversion will initially be attempted with above preferred VOSS image
- If conversion fails, all the other available VOSS images are tried consecutively
- Script never gives up, will continue trying with all the VOSS images available on USB
- Script only halts if USB stick is removed

USB ZTP execution



- When the Universal Hardware boots up in EXOS mode as a Switch Engine with factory default config AND a USB stick is inserted AND a “ztp.py” Python script is found on the USB stick, the Python script is executed
 - If the switch has already booted into EXOS and is still running an unsaved default config, the “ztp.py” script will also get executed upon USB stick insertion, without any need to manually unconfigure the switch into a fresh restart
 - On some older versions of EXOS seems the script gets executed twice, but that does not matter, one instance will succeed !
- If a console is connected, the following can be seen:

```
udevd: Starting version 243.2+
init started: BusyBox v1.28.3 (2020-10-25 20:59:38 America)
Starting ExtremeXOS 31.1.1b3
Copyright (C) 1996-2020 Extreme Networks. All rights reserved.
This product is protected by one or more US patents listed at https://www.extremenetworks.com/compa

System is in evaluation for 30 day(s) and this will expire in 29 day(s) (Wed Oct 5 08:36:49 2022)
(pending-AAA) login:

Authentication Service (AAA) on the master node is now available for login.

Saving configuration primary.cfg on master ..... done!
Saving configuration primary.cfg on master .. done!
USBZTP: usbztp version 1.0.0.6
USBZTP: file ztp.py
USBZTP: usbztp version 1.0.0.6
USBZTP: file ztp.py
Sent SIGTERM to all processesab
Sent SIGKILL to all processes
Requesting system reboot
```


ZTP log files – debug log file – unsuccessful conversions



- The Python script also maintains a debug log file on the switch internal flash
 - /usr/local/cfg/ztp-debug.log
- This file contains a log of all the CLI commands executed by the Python script and the output of those commands

```
(CIT_31.1.1.3) 5520-24W-EXOS.13 # cat ztp-debug.log
2022-09-05 11:18:50.179625# disable cli prompting

2022-09-05 11:18:50.181333# show version | include Switch
Switch          : 800992-00-AB SB032050G-00102 Rev AB BootROM: 2.2.1.8   IMG: 31.1.1.3

sendCLI_showRegex() raw data = ['SB032050G-00102']
2022-09-05 11:18:50.184376# show switch | include "System Type:"
System Type:      5520-24W-EXOS

sendCLI_showRegex() raw data = ['5520']
vossImageFiles = ['5320.8.7.0.0.voss', '5420.8.7.0.0.voss']
sortImageFiles() imageDict = {'5420': {'8.7.0.0': '5420.8.7.0.0.voss'}, '5320': {'8.7.0.0': '5320.8.7.0.0.voss'}}
sortImageFiles() orderedImages = ['5420.8.7.0.0.voss', '5320.8.7.0.0.voss']
2022-09-05 11:18:50.274259# download url file:///usr/local/ext/5420.8.7.0.0.voss install

Downloading to Switch.....
.....
Error: Failed to download image - File 5420.8.7.0.0.voss is not compatible with this platform.

(CIT_31.1.1.3) 5520-24W-EXOS.13 #
```

- Also included in this file are debug information of data extracted by the Python script
- This information is useful to the script author (myself!), if the script, for whatever reason, failed to convert a switch to Fabric Engine
- If the Fabric Engine conversion fails, the debug log file can be recovered from the Switch Engine internal flash

